

GANGHUA WANG

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EDUCATION

University of Minnesota, Twin Cities

Ph.D in Statistics, GPA: 4.0/4

Advised by [Prof. Jie Ding](#) and co-advised by [Prof. Yuhong Yang](#)

Aug. 2019 - present

Minneapolis, MN

Peking University

B.S. in Statistics, Minor in Economics, GPA: 3.7/4

Sept. 2015 - July 2019

Beijing, China

EXPERIENCE (SELECTED)

Cisco Research Fellow: Privacy-preserving machine learning

Jan. 2021 - present

Supervised by Prof. Jie Ding and Prof. Yuhong Yang, University of Minnesota

Collaborated with Cisco research team

Minneapolis, MN

We formalized the definition of model privacy and proposed model defense mechanisms that provide rigorous guarantees.

Research assistant: Data privacy

June 2020 - Aug. 2020

Supervised by Prof. Jie Ding, University of Minnesota

Minneapolis, MN

We proposed a novel framework to protect data privacy for each individual in the data collection procedure, which is different from differential privacy.

Summer internship: Component analysis in brain image

July. 2018 - Sept. 2018

Supervised by Prof. Hongyu Zhao, Yale University

New Haven, CT

We proposed to extract functional brain neural network components using the Latent Dirichlet Allocation method, which facilitates the understanding of the intrinsic structure of the human brain.

Research assistant: Community detection with co-variables

Dec. 2017 - June 2018

Supervised by Prof. Jinzhu Jia, Peking University

Beijing, China

We adopted a maximum likelihood estimation-based method to combine the information from both co-variables and the network to detect communities under the stochastic block model.

PUBLICATIONS

- [1] Ganghua Wang and Jie Ding. "Subset privacy: Draw from an obfuscated urn". *arXiv preprint*, 2021
- [2] Gen Li, Ganghua Wang, Yuantao Gu, and Jie Ding. "Provable identifiability of relu neural networks via lasso regularization". *arXiv preprint*, 2021
- [3] Ganghua Wang, Jie Ding, and Yuhong Yang. "Regression with set-valued categorical predictors". *arXiv preprint*, 2021

HONORS AND AWARDS (SELECTED)

Awarded by University of Minnesota, Twin Cities

School of Statistics First Year Scholarship

2019

Awarded by Peking University

The Academic Excellence Scholarship (3 times)

2016 - 2018

TECHNICAL STRENGTHS

Computer Languages

Python, MATLAB, R, SQL

Frameworks

Pytorch, TensorFlow, MPI